

PAW on-site valence exact exchange: CoQuí vs. the correct (all-electron) expression

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Abstract

This note documents (i) exactly what CoQuí currently computes for the PAW on-site valence exchange, term by term, from the source; (ii) what ABINIT computes and why it equals the true all-electron (AE) exchange; and (iii) a derivation of the difference between the two, which turns out to be a single, physically identifiable term that CoQuí neglects. The analysis is backed by the instrumented Si $a = 10.20$ comparison (both codes, finite-size correction off, on-site term toggled): CoQuí on-site exchange +0.0414 Ha vs. ABINIT +0.0925 Ha (converged), a 0.052 Ha gap that the derived missing term accounts for.

Source of record.

- CoQuí smooth + augmentation + one-center: `src/hamiltonian/paw/v_x_paw.hpp`
- CoQuí deltaC kernel: `src/python/mean_field/abinit_interface/paw_deltaC.py` (identical definition to QE's `pw2coqui`; stored in the mean-field h5 and read into `species_paw_t::deltaC`)
- ABINIT on-site oscillator: `abinit-10.6.7/src/65_paw/m_pawpwij.F90`, used by `calc_sigx_me (src/70_gw/m_sigx.F90)`

1 Notation

Per k -point pair (m, k_q) , (n, k_p) the *pair (transition) density* is $\rho_{mn}(\mathbf{r}) = \psi_{mk_q}^*(\mathbf{r}) \psi_{nk_p}(\mathbf{r})$. The exact-exchange energy is the bilinear

$$E_x = -\frac{1}{2} \sum_s \sum_{k_q k_p} \sum_{mn} \frac{f_{mk_q} f_{nk_p}}{N_k} \langle \rho_{mn} | v_C | \rho_{mn} \rangle, \quad \langle \rho | v_C | \rho' \rangle \equiv \frac{4\pi}{\Omega} \sum_G \frac{\rho^*(K) \rho'(K)}{|K|^2}, \quad K = G + k_p - k_q. \quad (1)$$

The matrix-element form actually built in the code is $K_{ij}(s, k_p) = -\frac{1}{N_k} \sum_{k_q} \sum_n f_{nk_q} \langle \rho_{ni} | v_C | \rho_{nj} \rangle$, with $E_x = \frac{1}{2} \sum_i f_i K_{ii}$.

PAW decomposition of the pair density. With smooth orbitals $\tilde{\psi}$, projectors p_I^a , and AE/PS partial waves $\phi_I^a, \tilde{\phi}_I^a$:

$$\rho_{mn}(\mathbf{r}) = \underbrace{\tilde{\rho}_{mn}(\mathbf{r})}_{\text{smooth (grid)}} + \sum_a \underbrace{[\rho_{mn,a}^1(\mathbf{r}) - \tilde{\rho}_{mn,a}^1(\mathbf{r})]}_{D_{mn,a}(\mathbf{r}) \text{ (on-site, radial)}}, \quad (2)$$

$$\rho_{mn,a}^1 = \sum_{IJ} P_{mI}^{a*} P_{nJ}^a \phi_I^a \phi_J^a, \quad \tilde{\rho}_{mn,a}^1 = \sum_{IJ} P_{mI}^{a*} P_{nJ}^a \tilde{\phi}_I^a \tilde{\phi}_J^a, \quad P_{nJ}^a \equiv \langle p_J^a | \tilde{\psi}_n \rangle. \quad (3)$$

The *compensation charge* $\hat{n}_{mn,a}$ reproduces the *multipole moments* of $D_{mn,a}$ but not its shape:

$$\hat{n}_{mn,a}(\mathbf{r}) = \sum_{IJ} P_{mI}^{a*} P_{nJ}^a \sum_L q_{IJ}^L g_L(r) \{Y_{LM}\}, \quad q_{IJ}^L = \int (\phi_I \phi_J - \tilde{\phi}_I \tilde{\phi}_J) r^L r^2 dr, \quad (4)$$

with g_L the (Bessel/Gaussian) shape function normalised to unit L -moment. By construction $\int (D_{mn,a} - \hat{n}_{mn,a}) r^L Y_{LM} d^3r = 0$ for all LM .

Two identities used below (Coulomb bilinearity + the moment property):

$$\langle D | v_C | D \rangle - \langle \rho^1 | v_C | \rho^1 \rangle + \langle \tilde{\rho}^1 | v_C | \tilde{\rho}^1 \rangle = -2 \langle \tilde{\rho}^1 | v_C | D \rangle \quad (\text{expand } D = \rho^1 - \tilde{\rho}^1). \quad (5)$$

2 What CoQuí currently implements

CoQuí builds K_{ij} as **three** additive pieces (`v_x_paw.hpp`, function `v_x`).

(1a) Smooth pair density (grid). Lines 442–446. For occupied n at k_q and every a at k_p :

$$\tilde{\rho}_{na}(G) = \text{FFT}[\tilde{u}_n^*(k_q, \mathbf{r}) \tilde{u}_a(k_p, \mathbf{r})](G). \quad (6)$$

(1b) Compensation-charge augmentation (grid), lines 448–485.

$$\rho_{na}^{\text{full}}(G) = \tilde{\rho}_{na}(G) + \Omega \sum_a \sum_{IJ} P_{nI}^{a*}(k_q) P_{aJ}^a(k_p) Q_a^{IJ}(K) e^{-iK \cdot \tau_a} \quad (7)$$

where $Q_a^{IJ}(K)$ is the Fourier transform of the **compensation charge** $\hat{n}_{IJ}$ (moments $\times$ shape function), carrying the $4\pi/\Omega$ prefactor (`evaluate_Q_IJ_at_K_fast`, l.229; the explicit Ω on l.461 restores the density normalisation). Crucially this is $\hat{n}$, i.e. **moments only** — it is *not* the AE–PS pair density. So on the grid CoQuí forms $\rho^{\text{full}} = \tilde{\rho} + \hat{n}$.

(1c) Coulomb contraction (smooth + augmentation), lines 487–497.

$$K_{ij}^{\text{sm+aug}}(s, k_p) = -\frac{1}{N_k \Omega} \sum_{k_q} \sum_n f_{nk_q} \sum_G v_C(K) \rho_{ni}^{\text{full}*}(G) \rho_{nj}^{\text{full}}(G), \quad \left[\text{scl} = -\frac{1}{N_k \Omega} \right]. \quad (8)$$

In bilinear form this is $-\frac{1}{N_k} \sum_{k_q} \sum_n f_n \langle \tilde{\rho} + \hat{n} | v_C | \tilde{\rho} + \hat{n} \rangle$.

(1d) One-center deltaC correction, lines 503–590.

$$\Delta K_{ij}^{\text{oc}}(s, k_p) = -\frac{1}{N_k} \sum_{k_q} \sum_n f_{nk_q} \sum_a \sum_{IJKL} P_{iI}^{a*}(k_p) P_{nJ}^a(k_q) \Delta C^a(I, J, K, L) P_{nK}^{a*}(k_q) P_{jL}^a(k_p), \quad (9)$$

with $\text{scl}_{\text{oc}} = -\frac{1}{N_k}$ and (`paw_deltaC.py`, l.111–142; QE/pw2coqui identical):

$$\Delta C^a(I, J, K, L) = K^{\text{AE}} - K^{\text{PS}}, \quad K = \sum_{LM} \int V_{LM}^{IJ}(r) \rho_{LM}^{KL}(r) dr, \quad (10)$$

$$\rho_{LM}^{IJ,AE} = \sum_{lm} \text{ap}(lm, l_i, l_j) \underbrace{\phi_I \phi_J}_{\text{pfunc}}, \quad (11)$$

$$\rho_{LM}^{IJ,PS} = \sum_{lm} \text{ap}(lm, l_i, l_j) \left[\underbrace{\tilde{\phi}_I \tilde{\phi}_J}_{\text{ptfunc}} + \underbrace{q_{IJ}^L g_L(r)}_{\text{qfunc}=\hat{n}} \right]. \quad (12)$$

Read literally: the AE side is the pure product $\phi_I \phi_J$; the PS side is $\tilde{\phi}_I \tilde{\phi}_J + \hat{n}_{IJ}$ (note the + `qfunc1` on l.107/126 of `paw_deltaC.py` and l.215 of `paw_onecenter.hpp`). Hence, per atom,

$$\Delta C = \langle \phi \phi | v_C | \phi \phi \rangle^{\text{AE}} - \left\langle \tilde{\phi} \tilde{\phi} + \hat{n} \middle| v_C \middle| \tilde{\phi} \tilde{\phi} + \hat{n} \right\rangle^{\text{PS}} = \langle \rho^1 | v_C | \rho^1 \rangle - \langle \tilde{\rho}^1 + \hat{n} | v_C | \tilde{\rho}^1 + \hat{n} \rangle. \quad (13)$$

(1e) CoQuí total (compensated PAW). Collecting (1c)+(1d) and cancelling the shared $\langle \hat{n} | v_C | \hat{n} \rangle$:

$$E_x^{\text{CoQuí}} = -\frac{1}{2} \sum \frac{ff}{N_k} \left[\langle \tilde{\rho} | v_C | \tilde{\rho} \rangle + 2 \langle \tilde{\rho} | v_C | \hat{n} \rangle - 2 \langle \tilde{\rho}^1 | v_C | \hat{n} \rangle + \langle \rho^1 | v_C | \rho^1 \rangle - \langle \tilde{\rho}^1 | v_C | \tilde{\rho}^1 \rangle \right] \quad (14)$$

This is the standard **compensated (moment-matched) PAW** exchange: the smooth density interacts with the *multipoles* $\hat{n}$ of the on-site AE–PS difference, plus an exact on-site AE–(PS+comp) Fock correction.

Prefactor note: `scl` = $-1/(N_k \Omega)$ for the grid Coulomb measure and `scloc` = $-1/N_k$ for the already-integrated radial `deltaC` are internally consistent and validated to machine precision against CoQuí’s own THC K_a (`test vx_onecenter_vs_thc_Ka`). The prefactors are **not** the issue here.

3 What ABINIT computes (= the correct AE exchange)

ABINIT builds the **oscillator** of the *full AE pair density* directly (`m_pawpwij.F90`). The on-site addition to $\langle \tilde{\psi}_m | e^{-iK\mathbf{r}} | \tilde{\psi}_n \rangle$ is (l.479, 527, 561, 859–918):

$$M_{mn}^{\text{onsite}}(K) = \sum_a \sum_{IJ} P_{mI}^{a*} P_{nJ}^a q_{IJ}^a(K), \quad q_{IJ}^a(K) = 4\pi \sum_{LM} (-i)^L Y_{LM}(\hat{K}) G_{l_i m_i, l_j m_j}^{LM} \text{ff}_{IJ}^{aL}(|K|) \quad (15)$$

$$\text{ff}_{IJ}^{aL}(K) = \int_0^{r_a} j_L(Kr) [\phi_I \phi_J - \tilde{\phi}_I \tilde{\phi}_J](r) dr \quad (16)$$

i.e. the transform of the **full AE–PS on-site pair density D , with the true radial shape and NO compensation charge** (`phiphj - tphitphj`, l.561; the `use_pawnhat0 = 0` default in `m_sigx.F90`:l.155). The exchange is then

$$E_x^{\text{ABINIT}} = -\frac{1}{2} \sum \frac{ff}{N_k} \langle \tilde{\rho} + D | v_C | \tilde{\rho} + D \rangle = -\frac{1}{2} \sum \frac{ff}{N_k} \left[\langle \tilde{\rho} | v_C | \tilde{\rho} \rangle + 2 \langle \tilde{\rho} | v_C | D \rangle + \langle D | v_C | D \rangle \right]. \quad (17)$$

Because $\tilde{\rho} + D$ is the exact AE pair density (up to the `ecutsigx` truncation of the sharp D , which we verified converged: on-site moves < 1 mHa from `ecutsigx` = 25 $\rightarrow$ 50), this equals the true AE exchange. This is the Arnaud–Alouani / Shishkin–Kresse PAW–GW oscillator.

4 The difference (derivation)

Subtract § (1e) from §3, per pair, using identity (5):

$$E_x^{\text{ABINIT}} - E_x^{\text{CoQuí}} = -\frac{1}{2} \sum \frac{f f}{N_k} \left[2 \langle \tilde{\rho} | v_C | D \rangle - 2 \langle \tilde{\rho} | v_C | \hat{n} \rangle + 2 \langle \tilde{\rho}^1 | v_C | \hat{n} \rangle - 2 \langle \tilde{\rho}^1 | v_C | D \rangle \right] \quad (18)$$

$$E_x^{\text{ABINIT}} - E_x^{\text{CoQuí}} = - \sum \frac{f_m f_n}{N_k} \langle \tilde{\rho}_{mn} - \tilde{\rho}_{mn}^1 | v_C | D_{mn} - \hat{n}_{mn} \rangle \quad (19)$$

This is the term CoQuí drops. Its two factors are each localised in/near the augmentation sphere:

- $\tilde{\rho} - \tilde{\rho}^1$ — the smooth pair density minus its on-site PS partial-wave expansion. Vanishes only if the PS partial waves are *complete* inside the sphere (the usual PAW assumption $\tilde{\rho}|_{\text{sphere}} = \tilde{\rho}^1$).
- $D - \hat{n}$ — the AE–PS on-site difference minus its multipole moments: the **shape of D beyond its moments**. Zero moments by construction, nonzero shape.

The compensated PAW total-energy formalism sets this cross term to zero. That is an excellent approximation for the *Hartree/total energy* (long-ranged, moment dominated) but **not** for **exchange**, which is short-ranged and sensitive to the pair-density shape inside the core. For Si the numbers are:

quantity (Si $a = 10.20$, $4 \times 4 \times 4$, finite-size off)	value (Ha)
smooth only, both codes (identical to 6 μHa)	-1.77964
ABINIT on-site contribution $2 \langle \tilde{\rho} v_C D \rangle + \langle D v_C D \rangle$	+0.0925
CoQuí $\langle \tilde{\rho} v_C \hat{n} \rangle$ -type augmentation (1b/1c extra)	+0.0481
CoQuí deltaC one-center (1d)	-0.0067
CoQuí on-site total	+0.0414
missing cross term $-\langle \tilde{\rho} - \tilde{\rho}^1 v_C D - \hat{n} \rangle$	$\approx$ +0.0511

The sign is consistent: CoQuí’s on-site is too small (recovers $\sim 44\%$), so its total exchange is too *negative* by 0.052 Ha (the missing term is repulsive/less-binding).

5 What I believe is the correct expression

Two mathematically equivalent routes recover the missing term. Both are “correct”; they differ in numerical convenience.

Route A — shape-restored oscillator (ABINIT/Arnaud–Alouani; recommended)

Replace CoQuí’s compensation-charge augmentation (§ 1b) with the **full AE–PS pair-density form factor**, and then **drop deltaC entirely**:

$$\rho_{na}^{\text{full}}(G) = \tilde{\rho}_{na}(G) + \sum_a \sum_{IJ} P_{nI}^{a*}(k_q) P_{aJ}^a(k_p) \mathcal{Q}_a^{IJ}(K) e^{-iK\tau_a}, \quad \mathcal{Q}_a^{IJ}(K) = \int_0^{r_a} j_L(Kr) [\phi_I \phi_J - \tilde{\phi}_I \tilde{\phi}_J] dr \times (\text{ang.}), \quad (20)$$

i.e. use `pfunc - ptfunc` (AE–PS product) in the radial Hankel transform *instead of* `qfunc1` (the compensation moments). Structurally this is a one-line change to `build_grad_tab / evaluate_Q_IJ_at_K`: feed `pfunc - ptfunc` rather than the `moment×shape` `qfunc1`. Then $K^{\text{sm+aug}}$ alone equals the exact AE exchange and the `deltaC` block (§ 1d) must be removed (it would double count).

- **Pro:** exact, matches ABINIT/VASP, no separate one-center kernel.
- **Con:** D is sharp; the augmentation FFT mesh (`fft_mesh_aug`) must resolve $\text{ff}_{IJ}^L(K)$ out to the same effective cutoff ABINIT uses (`ecutsigx`; here converged by ~ 50 Ha). This is the reason the compensated route was chosen originally — $\hat{n}$ is smooth and grid-friendly.

Route B — keep compensation, add the missing cross term to `deltaC`

Keep § 1b/1c as is, but redefine the one-center kernel so that it also supplies $-\langle \tilde{\rho} - \tilde{\rho}^1 | v_C | D - \hat{n} \rangle$. In practice this means the on-site Fock must be evaluated between the *smooth-minus-onsite* density and the *shape residual* $D - \hat{n}$, not only the pure on-site AE/PS products. This is harder to cast as a frozen n_h^4 matrix because $\tilde{\rho} - \tilde{\rho}^1$ depends on the actual smooth orbitals, not just the projector amplitudes — so Route A is cleaner.

Recommendation

Implement **Route A**: swap the augmentation form factor from the compensation charge (`qfunc1`, moments) to the full AE–PS partial-wave product (`pfunc - ptfunc`, shape), on a mesh converged like ABINIT’s `ecutsigx`, and drop `deltaC` from the exchange. Validate against the numbers here (Si on-site must go $+0.0414 \rightarrow +0.0925$ Ha; total vv exchange $-1.7382 \rightarrow -1.6863$ Ha; per-band Σ_x must match ABINIT `iszoff` band-for-band). Keep the compensation-charge path for the **Hartree** `deeq` (where it is correct and grid-friendly).

Caveat / next check. The identity in §4 assumes the exchange is the clean bilinear $\langle \rho_{mn} | v_C | \rho_{mn} \rangle$ and that CoQu’s terms map exactly as written above; the 0.052 Ha numerical closure supports this. Before implementing, reproduce the decomposition at a second system (e.g. LiH or a 3d oxide, where on-site exchange is larger) to confirm the missing-term identity holds beyond Si *sp*.

6 Implementation status and open items (update)

Route A was implemented on the `paw` branch. This section records what was done, what is verified, and the two issues that remain before it is production-ready and numerically confirmed against ABINIT.

What was implemented (committed)

- `build_grad_tab_full_aeps (paw_aug_q_eval.hpp)`: Hankel transform of the *full* AE–PS pair density $D_{ijv}(r) = \phi_{n_b} \phi_{m_b} - \tilde{\phi}_{n_b} \tilde{\phi}_{m_b}$ per β pair, reusing the same ijv/L layout as the compensation table so it drops into `evaluate_Q_IJ_at_K_fast` unchanged.
- `v_x_paw.hpp`: **both** the diagonal (*nii*) and full density-matrix (*nij*, GW natural-orbital) exchange kernels take a `shape_restored` flag; when set, PAW species use the full-AE–PS table and the `deltaC` block is skipped. USPP is unaffected.
- `pseudopot::set_paw_exx_shape_restored()` threads the flag through `add_exchange`. **Default off**, so the compensated path and all existing direct-vs-THC tests are unchanged.

- An env-gated (COQUI_PAW_SHAPE_RESTORED) analogue in the THC augmentation, and env-gated direct-exchange diagnostics in `rpa.cpp`.

What is verified

The hamiltonian unit test `vexchange_shape_restored` passes on a PAW fixture: Option A genuinely changes the exchange ($\max |\Delta K| = 5.6 \times 10^{-4}$), and the **full density-matrix (nij) path equals the diagonal (nii) path to 3.3×10^{-16}** (machine precision) under Option A. This validates the implementation and the GW density-matrix handling.

Open item 1 — the augmentation grid (THC vs direct)

The sharp AE–PS pair density D must be resolved on the *dense* grid. For Si $a = 10.20$ the smooth grid is 18^3 (~ 60 Ry) and the dense grid is 48^3 ($\text{ecutrho} = 121$ a.u., comparable to ABINIT’s `ecutsigx`).

- The **THC** augmentation runs on the *smooth* grid, so with the full-AE–PS form factor it captures only $\sim 6\%$ of the on-site correction: $\text{vv exchange } -1.7382 \rightarrow -1.7350$ (measured), vs. the dense-grid target -1.6863 . The compensation charge worked on the smooth grid only because it is smooth by construction.
- The **direct** `v_x.paw` already uses the dense grid (`fft_mesh_aug`) and is therefore the correct-grid path.

Consequence: a production THC Option A needs the augmentation migrated to the dense grid; the direct path is correct today via `set_paw_exx_shape_restored(true)`.

Open item 2 — the direct path is too slow (why, vs ABINIT)

The direct `v_x` is correct but, as written, intractable for meshes with many k -points (it timed out at every configuration on Si $a = 10.20$, and even on Si $2 \times 2 \times 2$). ABINIT computes the same physics in seconds. The difference is purely implementation:

ABINIT (<code>m_pawpwij.F90</code>)	CoQuí direct <code>v_x</code>
Radial form-factor spline built once over a <i>bounded</i> q -grid ($\leq \text{ecutsigx} \approx 10$ a.u.)	grad Bessel table rebuilt every <code>v_x</code> call , over an oversized K -range
K -range $\sim \text{ecutsigx}$ sphere	<code>K_max_est</code> = abs-sum of dense FFT-box corners ≈ 77 a.u. ($n_K \approx 7700$; $\sim 4 \times 10^7$ <code>sph.bessel</code> evals per table)
Works over the compact plane-wave sphere <code>npwx</code> ≈ 1591 – 4477	Works over the <i>full</i> dense FFT grid <code>nnr</code> = 110592 (incl. box corners outside the physical sphere)
One vectorized <code>splfit</code> over all <code>npwx</code> per (IJ, L) ; no allocation	scalar evaluation per (g, I, J)

Empirically the run hangs in the `v_x` setup phase, consistent with the oversized per-call grad-table build dominating.

Fix (planned/in progress). (1) Bound `k_max` to the physical cutoff (true grid-corner max / `ecutrho` sphere) instead of the abs-sum; (2) restrict the augmentation build + Coulomb sum to the G -sphere (`~ecutsigx`), not the full FFT box; (3) build the form-factor table *once* (cache across the two passes) and vectorize the radial evaluation (no per- (g, I, J) allocation). With these, the direct path should match ABINIT's cost and finally allow the number-vs-number confirmation (Si $2 \times 2 \times 2$ target: shape-restored = ABINIT full-onsite -1.316447 Ha; $a = 10.20$: -1.6863 Ha).

Confirmation status. **Not yet numerically confirmed against ABINIT.** The nij /density-matrix correctness and the by-construction equivalence are established; the code-vs-code energy match awaits the direct-path speedup above (or a dense-grid THC augmentation).